



Factoring Polynomials Completely

Every polynomial of degree $n > 0$ can be written as the product of n linear factors.

Factoring a Binomial Over the Set of Complex Numbers

EXAMPLE 1

Write $x^2 + 25$ as the product of two linear factors.

SOLUTION

The equation $x^2 + 25 = 0$ is equivalent to the equation $x^2 = -25$, which has solutions $5i$ and $-5i$. Therefore, $x^2 + 25 = (x + 5i)(x - 5i)$. ■

The polynomial in Example 1 has no real zeros, but it has two complex zeros. Some polynomials have both real zeros and complex zeros.

► **Think About It** Recall that the set of real numbers is a subset of the set of complex numbers. If a polynomial has both real zeros and complex zeros, then all of its zeros are complex. In fact, all the zeros of a polynomial can be considered complex.

Understanding Multiplicity

For a polynomial that has a linear factor appearing more than once, the idea of multiplicity is useful.

Definition

For any polynomial $p(x)$, a root a of $p(x)$ has **multiplicity** m if the factor $(x - a)$ occurs m times in the factorization of $p(x)$.

If a is a real root with odd multiplicity, the graph of the function crosses the x -axis at $x = a$. If a is a real root with even multiplicity, the graph touches, but does not cross, the x -axis at $x = a$.

EXAMPLE 2

Given that $f(x) = x^4 - 5x^3 + x^2 + 21x - 18 = (x - 3)(x - 3)(x - 1)(x + 2)$:

A Find the multiplicity of each root of $f(x)$.

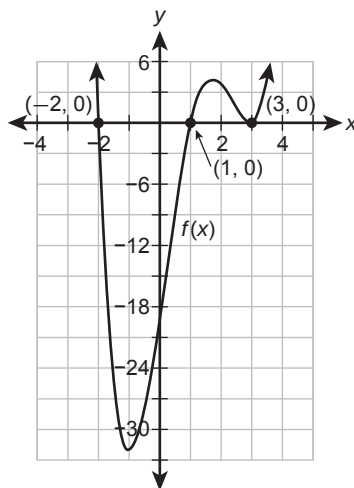
SOLUTION

The roots are 3, 3, 1, and -2 . The root 3 occurs two times, so it has a multiplicity of 2. The roots 1 and -2 occur once, so each has a multiplicity of 1.

B Describe the behavior of the graph at each root.

SOLUTION

The graph of $f(x) = x^4 - 5x^3 + x^2 + 21x - 18$ shows the behavior of the function at its zeros. The graph crosses the x -axis at the zeros of odd multiplicity: $x = 1$ and $x = -2$. The graph touches, but does not cross, the x -axis at the root of even multiplicity: $x = 3$.



Finding the Number of Roots of a Polynomial

Fundamental Theorem of Algebra

Every polynomial $p(x)$ with degree n , $n > 0$, has at least one complex root.

Corollary Every polynomial $p(x)$ with degree n , $n > 0$, has n complex roots as long as any root with multiplicity m is counted m times.

EXAMPLE 3

Find the number of complex roots of the polynomial.

A $x^5 + 3x^4 + x^2 + 1$

SOLUTION

The polynomial has degree 5, so there are five complex roots.

B $x - 3$

SOLUTION

The polynomial has degree 1, so there is one complex root. ■

Factoring a Polynomial Over the Set of Complex Numbers

How to Factor a Polynomial Completely Over the Set of Complex Numbers

Step 1 Use the degree of the polynomial to determine the total number of complex roots you need to find.

Step 2 Identify the possible rational roots.

Step 3 Find rational roots (using a graph, synthetic division, or other strategies) and use them to factor the polynomial.

Step 4 Factor the remaining polynomial.

EXAMPLE 4

Factor the polynomial completely. Then sketch the graph.

A $f(x) = x^3 - 6x^2 + 10x - 8$

SOLUTION

Step 1 The polynomial is of degree 3, so there are three complex roots.

Step 2 Use the rational root theorem to identify all possible rational roots. The leading coefficient is 1 and the constant is -8 , so the possible rational roots are ± 1 , ± 2 , ± 4 , and ± 8 .

Step 3 Of the possible rational roots, only 4 is actually a root.

	1	-6	10	-8
1	1	-5	5	-3
-1	1	-7	17	-25
2	1	-4	2	-4
-2	1	-8	26	-60
4	1	-2	2	0

$$\text{So } f(x) = (x - 4)(1x^2 - 2x + 2) = (x - 4)(x^2 - 2x + 2).$$

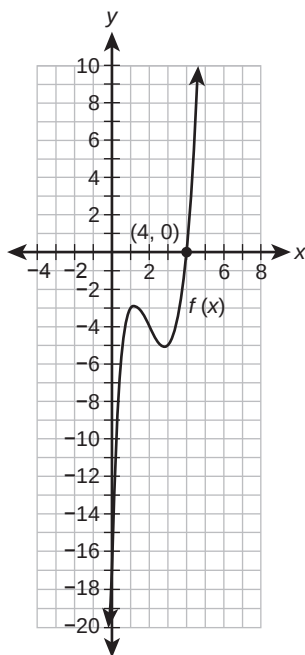
► **Remember** All real numbers are complex numbers of the form $a + bi$, where $b = 0$. The real root 4 is also the complex root $4 + 0i$.

Step 4 Find the two remaining roots. Use the quadratic formula to solve $x^2 - 2x + 2 = 0$.

$$\begin{aligned} x &= \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1} = \frac{2 \pm \sqrt{-4}}{2} \\ &= \frac{2 \pm 2i}{2} \\ &= 1 \pm i \end{aligned}$$

The roots $1 + i$ and $1 - i$ are complex. So the complete factorization of $f(x)$ is $f(x) = (x - 4)[x - (1 + i)][x - (1 - i)]$.

The roots of the polynomial $f(x) = x^3 - 6x^2 + 10x - 8$ are 4, $1 + i$, and $1 - i$. In the graph of a function with both real roots and complex roots, only the real roots are x -intercepts. The graph of $f(x)$ shows that only the real root, 4, is an x -intercept.



B $f(x) = x^4 - 2x^3 - 3x^2 + 2x + 2$

SOLUTION

Step 1 The polynomial is of degree 4, so there are four complex roots.

Step 2 By the rational root theorem, the possible rational roots are ± 1 and ± 2 .

Step 3 Synthetic division shows -1 and 1 as roots.

$$\begin{array}{r|rrrrrr} -1 & 1 & -2 & -3 & 2 & 2 \\ & & -1 & 3 & 0 & -2 \\ \hline & 1 & -3 & 0 & 2 & 0 \end{array}$$

$$\begin{array}{r|rrrr} 1 & 1 & -3 & 0 & 2 \\ & & 1 & -2 & -2 \\ \hline & 1 & -2 & -2 & 0 \end{array}$$

So $f(x) = (x+1)(x-1)(1x^2 - 2x - 2)$.

► **Think About It** Use the quotient from the first synthetic division to set up the synthetic division for the next root. This will save you some work and give you a quotient that is a quadratic.

Step 4 Find the two remaining roots. Use the quadratic formula to solve $x^2 - 2x - 2 = 0$.

$$\begin{aligned} x &= \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1(-2)}}{2 \cdot 1} = \frac{2 \pm \sqrt{12}}{2} \\ &= \frac{2 \pm 2\sqrt{3}}{2} \\ &= 1 \pm \sqrt{3} \end{aligned}$$

The roots $1 + \sqrt{3}$ and $1 - \sqrt{3}$ are irrational.

The factorization of $f(x)$ is

$$f(x) = (x+1)(x-1)\left[x - (1 + \sqrt{3})\right]\left[x - (1 - \sqrt{3})\right].$$

Because all the roots are real, all four roots are x -intercepts, as shown in the graph. The irrational roots are approximately -0.73 and 2.73 .

